ORF 524 – Final Exam

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Contents

Notes:

- Duration of examination: *4 hours*.
  An additional 20 minutes are given to upload your solutions to Canvas.
- All parts are equally weighted (5 points).
  Please provide as much detail as possible in your answers.
  Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
  A single PDF file must be submitted to Canvas.
- Only two pages of personal notes are allowed during the examination.
  Basic auxiliary information about standard distributions is provided at the end of the exam.
  This is otherwise a closed-book and individual examination.
  Do not consult textbooks, online materials, or other sources.
  Do not consult with classmates, other people or in online forums.

1 Question 1: V-statistics and U-statistics (25 points)

Let $(X_i : 1 \le i \le n)$ be an i.i.d. random sample with $\mathbb{E}[X_i] = \mu$ and $\mathbb{V}[X_i] = 1$. All limits are taken as $n \to \infty$, and $\rightsquigarrow$ denotes convergence in distribution.

A *V-statistic* is a generalization of the sample average to multiple sums over more than one observation at a time. For example, consider the following simple V-statistic:

$$\bar{V}_n = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} X_i X_j = \left(\frac{1}{n} \sum_{i=1}^{n} X_i \right)^2 .$$

A *U-statistic* is a V-statistic which excludes the diagonal terms and is properly rescaled. In the above example, we have

$$\bar{U}_n = \frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} X_i X_j = \binom{n}{2}^{-1} \sum_{1 \le i < j \le n} X_i X_j, \qquad \binom{n}{2} = \frac{n!}{2!(n-2)!}.$$

The *Hoeffding decomposition* gives a very useful representation for U-statistics (and hence also V-statistics). For the above example, it gives

$$\bar{U}_n = \mu^2 + \bar{L}_n + \bar{Q}_n,$$

where

$$\bar{L}_n = \frac{1}{n} \sum_{i=1}^{n} L(X_i), \qquad L(X_i) = 2\mu\big(X_i - \mu\big),$$

$$\bar{Q}_n = \binom{n}{2}^{-1} \sum_{1 \le i < j \le n} Q(X_i, X_j), \qquad Q(X_i, X_j) = (X_i - \mu)(X_j - \mu).$$

1. Show that $\mathbb{E}[\bar{L}_n] = 0$, $\mathbb{E}[\bar{Q}_n] = 0$ and $\mathbb{E}[\bar{L}_n \bar{Q}_n] = 0$.
   Show that $\mathbb{V}[\bar{U}_n] = \mathbb{V}[\bar{L}_n] + \mathbb{V}[\bar{Q}_n]$.

   - **Solution**: $\mathbb{E}[\bar{L}_n] = 0$ since $\mathbb{E}[X_i] = \mu$ and by linearity of expectation.
     $\mathbb{E}[\bar{Q}_n] = 0$ since X_i are independent and so $\mathbb{E}[(X_i - \mu)(X_j - \mu)] = 0$ for $i \neq j$.
     $\mathbb{E}[\bar{L}_n \bar{Q}_n] = 0$ because it is a sum of terms of the form $\mathbb{E}[(X_i - \mu)(X_j - \mu)(X_k - \mu)]$ where $i \neq j$. Therefore either $k \neq i$ or $k \neq j$ so this reduces to $\mathbb{E}[X_i - \mu]\mathbb{E}[(X_j - \mu)(X_k - \mu)] = 0$ or $\mathbb{E}[X_j - \mu]\mathbb{E}[(X_i - \mu)(X_k - \mu)] = 0$ by independence of X_i.

2. Show that $\mathbb{V}[\bar{L}_n] = O(n^{-1})$ and $\mathbb{V}[\bar{Q}_n] = O(n^{-2})$.
   Show that $\bar{L}_n = O_{\mathbb{P}}(n^{-1/2})$ and $\bar{Q}_n = O_{\mathbb{P}}(n^{-1})$.

   - **Solution**: $\mathbb{V}[\bar{L}_n] = \frac{1}{n^2} \sum_{i=1}^{n} \mathbb{V}[2\mu(X_i - \mu)] = \frac{4\mu^2}{n}$.

$$\mathbb{V}[\bar{Q}_n] = \binom{n}{2}^{-2} \mathbb{E}\Big[\Big(\sum_{1\le i<j\le n} (X_i-\mu)(X_j-\mu)\Big)^2\Big].$$

Expand the terms in the expectation, and then we can find: $\mathbb{E}[(X_i-\mu)^2(X_j-\mu)^2] = 1$ and there are $\binom{n}{2}$ such terms; the expected values of all other terms are 0 by independence. Then the second result follows.

The other two conclusions directly follow from Markov's inequality.

3. Show that $\bar{V}_n = \bar{U}_n + O_{\mathbb{P}}(n^{-1})$.

 - **Solution**: Note that

$$\bar{V}_n = \frac{n-1}{n}\bar{U}_n + \frac{1}{n^2}\sum_{i=1}^n X_i^2 = \bar{U}_n - \frac{1}{n}\bar{U}_n + \frac{1}{n}\Big(\frac{1}{n}\sum_{i=1}^n X_i^2\Big),$$

 and the result follows because $\bar{U}_n = \mu^2 + O_{\mathbb{P}}(n^{-1/2}) + O_{\mathbb{P}}(n^{-1}) = O_{\mathbb{P}}(1)$, using the Hoeffding decomposition $\bar{U}_n = \mu^2 + \bar{L}_n + \bar{Q}_n$, and $\frac{1}{n}\sum_{i=1}^n X_i^2 = O_{\mathbb{P}}(1)$, using the LLN.

4. Show that if $\mu \neq 0$ then $\sqrt{n}(\bar{U}_n - \mu^2) \rightsquigarrow \mathsf{N}(0, 4\mu^2)$.

 - **Solution**: We have

$$\begin{aligned}\sqrt{n}(\bar{U}_n - \mu^2) &= \sqrt{n}\bar{L}_n + \sqrt{n}\bar{Q}_n = \sqrt{n}\bar{L}_n + o_{\mathbb{P}}(1)\\ &= \frac{1}{\sqrt{n}}\sum_{i=1}^n L(X_i) + o_{\mathbb{P}}(1),\end{aligned}$$

 and the result follows by the CLT because $\mathbb{E}[L(X_i)] = 0$ and $\mathbb{V}[L(X_i)] = 4\mu^2$.

5. Show that if $\mu = 0$ then $n\bar{U}_n \rightsquigarrow \chi_1^2 - 1$.

 - **Solution**: Using the relation between U- and V-statistics, LLN, CLT and CMT, it immediately follows that

$$\begin{aligned}n\bar{U}_n &= \frac{n^2}{n-1}\bar{V}_n - \frac{1}{n-1}\sum_{i=1}^n X_i^2\\ &= \frac{n^2}{n-1}\left(\frac{1}{n}\sum_{i=1}^n X_i\right)^2 - \frac{1}{n-1}\sum_{i=1}^n X_i^2\\ &\rightsquigarrow \chi_1^2 - 1.\end{aligned}$$

2 Question 2: Uniform Inference at a Boundary (35 points)

Let $(X_i : 1 \leq i \leq n)$ be an i.i.d. random sample from $X_i \sim \mathbb{P}_\theta \in \mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta\}$, where X_i is real-valued and $\Theta = [0, \infty)$. Suppose $\mathbb{E}_\theta[X_i] = \theta$ and $\mathbb{V}_\theta[X_i] = 1$ for all $\theta \in \Theta$. Consider the estimator:

$$\hat{\theta}_n = \max\{\bar{X}_n, 0\}.$$

1. Suppose $\mathbb{P}_\theta = \mathsf{N}(\theta, 1)$. Show that $\hat{\theta}_n$ is the maximum likelihood estimator for θ.

   - **Solution**: follows directly by constrained optimization of the log-likelihood function.

2. Show that $\hat{\theta}_n \to_{\mathbb{P}} \theta$ for all $\theta \in \Theta$.

   - **Solution**: since the sequence is i.i.d., by the LLN $\bar{X}_n \to_{\mathbb{P}} \theta$ and because $x \mapsto \max\{x, 0\}$ is a continuous function, by the continuous mapping theorem,

   $$\hat{\theta}_n = \max\{\bar{X}_n, 0\} \to_{\mathbb{P}} \max\{\theta, 0\} = \theta,$$

   because $\theta \in \Theta = [0, \infty)$.

3. Suppose $\theta > 0$. Show that $\sqrt{n}(\hat{\theta}_n - \theta) \rightsquigarrow \mathsf{N}(0, 1)$.

   - **Solution**: first note that

   $$\sqrt{n}(\hat{\theta}_n - \theta) = \sqrt{n}(\max\{\bar{X}_n, 0\} - \theta) = \max\left\{\sqrt{n}(\bar{X}_n - \theta), -\sqrt{n}\theta\right\}.$$

   Now, for the first term inside the $\max\{\cdot\}$, we have a centered and scaled sample average of n i.i.d. r.v.'s with $\mathbb{E}_\theta[X_i] = \theta$ and $\mathbb{V}_\theta[X_i] = 1$, and therefore by the CLT we have $\sqrt{n}(\bar{X}_n - \theta) \rightsquigarrow \mathsf{N}(0, 1)$. On the other hand, since $\theta > 0$, $-\sqrt{n}\theta \to -\infty$. Since $\sqrt{n}(\bar{X}_n - \theta) = O_{\mathbb{P}}(1)$, while the other term is going to $-\infty$, so for n large enough,

   $$\max\left\{\sqrt{n}(\bar{X}_n - \theta), -\sqrt{n}\theta\right\} = \sqrt{n}(\bar{X}_n - \theta) + o_{\mathbb{P}}(1) \rightsquigarrow \mathsf{N}(0, 1),$$

   which gives the result.

4. Suppose $\theta = 0$. Show that $\sqrt{n}(\hat{\theta}_n - \theta) \rightsquigarrow \max\{\mathsf{N}(0, 1), 0\} =: \mathcal{L}(0)$.
   Give the exact form of the c.d.f. of the limiting law $\mathcal{L}(0)$.

   - **Solution**: we have $\sqrt{n}\hat{\theta}_n = \sqrt{n}\max\{\bar{X}_n, 0\} = \max\{\sqrt{n}\bar{X}_n, 0\}$. The first term inside the $\max\{\cdot\}$ is a mean-zero scaled sample average of n i.i.d. r.v.'s because $\mathbb{E}_\theta[X_i] = \theta = 0$ and $\mathbb{V}_\theta[X_i] = 1$. Thus, by the CLT, $\sqrt{n}\bar{X}_n \rightsquigarrow \mathsf{N}(0, 1)$, and therefore by the continuous mapping theorem

   $$\sqrt{n}\hat{\theta}_n = \max\{\sqrt{n}\bar{X}_n, 0\} \rightsquigarrow \max\{\mathsf{N}(0, 1), 0\} := Y.$$

Define $Z \sim \mathsf{N}(0,1)$. Then,

$$\mathbb{P}(Y \leq y) = \mathbb{P}(\max\{Z, 0\} \leq y) = \begin{cases} \Phi(y) & \text{if } y \geq 0 \\ 0 & \text{if } y < 0 \end{cases},$$

which completes the derivation.

5. Suppose $\mathbb{P}_\theta = \mathbb{P}_{\theta_n}$ with $\theta_n = \rho/\sqrt{n}$ for some $\rho > 0$.
Providing appropriate regularity conditions, show that $\sqrt{n}(\hat{\theta}_n - \theta_n) \rightsquigarrow \mathcal{L}(\rho)$.
Give the exact form of the c.d.f. of the limiting law $\mathcal{L}(\rho)$.
Hint: Recall the triangular array CLT.

   - **Solution**: By the hint, recall that if $(X_{n,i} : 1 \leq i \leq n)$ are independent random variables for each $n \geq 1$ with $\mathbb{E}[X_{n,i}] = 0$ and $\mathbb{E}[|X_{n,i}|^{2+\delta}] < \infty$ for some $\delta > 0$ (the so-called Lyapunov condition), then $\frac{1}{\sigma\sqrt{n}} \sum_{i=1}^{n} X_{n,i} \rightsquigarrow \mathsf{N}(0,1)$ as $n \to \infty$.
   Therefore, we conclude that if $\mathbb{E}_{\theta_n}[|X_{n,i}|^{2+\delta}] < \infty$ for some $\delta > 0$, then $\sqrt{n}(\bar{X}_n - \theta_n) \rightsquigarrow \mathsf{N}(0,1)$. By CMT,

   $$\sqrt{n}(\hat{\theta}_n - \theta_n) = \max\left\{\sqrt{n}(\bar{X}_n - \theta_n), -\sqrt{n}\theta_n\right\} \rightsquigarrow \max\{\mathsf{N}(0,1), -\rho\} := Y(\rho).$$

   Then,

   $$\mathbb{P}(Y(\rho) \leq y) = \mathbb{P}(\max\{Z, -\rho\} \leq y) = \begin{cases} \Phi(y) & \text{if } y \geq -\rho \\ 0 & \text{if } y < -\rho \end{cases},$$

   where $Z \sim \mathsf{N}(0,1)$.

6. Consider the following hypothesis testing problem for some $\theta_0 > 0$:

$$\mathsf{H}_0 : \theta = \theta_0 \qquad \text{vs.} \qquad \mathsf{H}_1 : \theta \neq \theta_0.$$

The Wald statistic is $T_n(\theta_0) = \sqrt{n}(\hat{\theta}_n - \theta_0)$, which leads to the rejection rule (under H_0):

$$\text{Reject } \mathsf{H}_0 \iff |T_n(\theta_0)| > \Phi^{-1}(1 - \alpha/2),$$

where $\Phi(x)$ denotes the c.d.f. of the standard Gaussian distribution, and $\alpha \in (0,1)$.

   (a) Show that this test is a pointwise α-size test asymptotically. That is, show that

   $$\sup_{\theta_0 > 0} \limsup_{n \to \infty} \mathbb{P}_{\theta_0}\left[|T_n(\theta_0)| > \Phi^{-1}(1 - \alpha/2)\right] = \alpha.$$

      - **Solution**: follows from the pointwise result in part 3 above, replacing lim by lim sup, and then taking the supremum over $\theta_0 > 0$.

(b) Show that this test is a uniform α-size test asymptotically. That is, show that

$$\limsup_{n\to\infty} \sup_{\theta_0>0} \mathbb{P}_{\theta_0}\left[|T_n(\theta_0)| > \Phi^{-1}(1-\alpha/2)\right] = \alpha.$$

- **Solution**: Note that for all $a > 0$ and $x \in \mathbb{R}$ we have $|\max\{x, -a\}| \le |x|$. So for any $\theta_n > 0$,

$$\begin{aligned}
\mathbb{P}_{\theta_n}\left[|T_n(\theta_n)| > \Phi^{-1}(1-\alpha/2)\right] &= \mathbb{P}_{\theta_n}\left[|\max\{\sqrt{n}(\bar{X}_n - \theta_n), -\sqrt{n}\theta_n\}| > \Phi^{-1}(1-\alpha/2)\right] \\
&\le \mathbb{P}_{\theta_n}\left[|\sqrt{n}(\bar{X}_n - \theta_n)| > \Phi^{-1}(1-\alpha/2)\right] \\
&\to \mathbb{P}\left[|\mathsf{N}(0,1)| > \Phi^{-1}(1-\alpha/2)\right] \\
&= \alpha.
\end{aligned}$$

Now take $\sup_{\theta_n>0}$ and $\limsup_{n\to\infty}$. The lower bound follows from part (a).

3 Question 3: Large Sample Estimation and Inference (40 points)

Let $(X_i : 1 \leq i \leq n)$ be i.i.d. with Lebesgue density $f(x-\theta)$ for some $\theta \in \Theta \subseteq \mathbb{R}$, where $f : \mathbb{R} \to \mathbb{R}$ is uniformly continuous and satisfies $f(x) = f(-x)$ and $f(0) > 0$. Suppose that n is odd and let $\hat{\theta}_{(1/2)}$ be the sample median. It was shown in one of the problem sets that

$$\sqrt{n}\big(\hat{\theta}_{(1/2)} - \theta\big) \rightsquigarrow \mathsf{N}\left(0, \frac{1}{4f(0)^2}\right).$$

1. Show that the population median of X_i is unique and equal to θ.
   Show that $\hat{\theta}_{(1/2)}$ is unique and consistent for θ.
   Assuming that $\mathbb{E}[\hat{\theta}_{(1/2)}]$ exists, is $\hat{\theta}_{(1/2)}$ unbiased for θ?

   - **Solution**: Note θ is a median because $\mathbb{P}(X_i \leq \theta) = \int_{-\infty}^{\theta} f(x-\theta)\,\mathrm{d}x = \int_{-\infty}^{0} f(x)\,\mathrm{d}x = 1/2$ since $f(x) = f(-x)$ and $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$. Also $f(0) > 0$ and f is continuous so it is positive on a neighborhood of zero, so for any $x < \theta$ we have $\mathbb{P}(X_i \leq x) < 1/2$ and for any $x > \theta$ we have $\mathbb{P}(X_i \leq x) > 1/2$.

     The estimator $\hat{\theta}_{(1/2)}$ is unique because n is odd.

     We have $\hat{\theta}_{(1/2)} - \theta = O_{\mathbb{P}}(n^{-1/2})$ and hence $\hat{\theta}_{(1/2)} = \theta + o_{\mathbb{P}}(1)$.

     By symmetry, $X \sim 2\theta - X$, which implies $\mathbb{E}[\hat{\theta}_{(1/2)}] = 2\theta - \mathbb{E}[\hat{\theta}_{(1/2)}]$ whenever the expectation exists.

2. Assuming sufficient regularity conditions, find the log-likelihood, score and Hessian functions, denoted by $\ell_n(\theta)$, $\ell_{\theta,n}(\theta)$ and $\ell_{\theta\theta,n}(\theta)$, respectively.
   Find the Cramér-Rao lower bound for unbiased estimators of θ, denoted by $\mathsf{CRLB}(\theta)$.
   Does $\mathsf{CRLB}(\theta)$ depend on θ?

   - **Solution**: Assuming $f(\cdot)$ is twice continuously differentiable (and integration and differentiation can be interchanged), we have

     $$\ell_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(X_i - \theta),$$
     $$\ell_{\theta,n}(\theta) = \frac{\partial}{\partial\theta}\ell_n(\theta) = -\frac{1}{n}\sum_{i=1}^{n} \frac{f'(X_i-\theta)}{f(X_i-\theta)},$$
     $$\ell_{\theta\theta,n}(\theta) = \frac{\partial^2}{\partial\theta^2}\ell_\theta(\theta) = \frac{1}{n}\sum_{i=1}^{n} \frac{f''(X_i-\theta)f(X_i-\theta) - f'(X_i-\theta)^2}{f(X_i-\theta)^2},$$

     Then, using the symmetry property, we have $\mathsf{CRLB}(\theta) = (n\mathcal{I}(\theta))^{-1}$ with

     $$\mathcal{I}(\theta) = \mathbb{E}\big[\ell_\theta(\theta)^2\big] = \int_{\mathbb{R}} \frac{f'(x)^2}{f(x)}dx,$$

     provided the integral exists. Clearly, the bound does not depend on θ.

3. Suppose that $\tilde{\theta}$ is a $\sqrt{n}$-consistent estimator of θ. Consider the following one-step estimator:

$$\hat{\theta}_{\texttt{1step}} = \tilde{\theta} - \frac{\ell_{\theta,n}(\tilde{\theta})}{\ell_{\theta\theta,n}(\tilde{\theta})}$$

Providing appropriate regularity conditions, show that $\sqrt{n}(\hat{\theta}_{\texttt{1step}} - \theta)$ has an asymptotic normal distribution, and that it is asymptotically efficient.
Hint: Expand $\ell_{\theta,n}(\tilde{\theta})$ around $\ell_{\theta,n}(\theta)$.

- **Solution**: First write

$$\sqrt{n}(\hat{\theta}_{\texttt{1step}} - \theta) = \sqrt{n}\Big(\tilde{\theta} - \theta - \frac{\ell_{\theta,n}(\tilde{\theta})}{\ell_{\theta\theta,n}(\tilde{\theta})}\Big).$$

By Taylor expansion, we have $\ell_{\theta,n}(\tilde{\theta}) = \ell_{\theta,n}(\theta) + \ell_{\theta\theta,n}(\check{\theta})(\tilde{\theta} - \theta)$ for some $\check{\theta}$ between θ and $\tilde{\theta}$. Then it follows that

$$\sqrt{n}(\hat{\theta}_{\texttt{1step}} - \theta) = -\frac{\sqrt{n}\ell_{\theta,n}(\theta)}{\ell_{\theta\theta,n}(\check{\theta})} + \sqrt{n}\ell_{\theta,n}(\tilde{\theta})\left(\frac{1}{\ell_{\theta\theta,n}(\check{\theta})} - \frac{1}{\ell_{\theta\theta,n}(\tilde{\theta})}\right).$$

Both $\tilde{\theta}$ and $\check{\theta}$ are consistent. Under regularity conditions, the second term is $o_{\mathbb{P}}(1)$, and by CLT and properties of score and information matrix, the first term is asymptotically normal with asymptotic variance equal to $\mathsf{CRLB}(\theta)$. So it is asymptotically efficient.

4. Suppose that there exists a well-defined maximum likelihood estimator $\hat{\theta}_{\texttt{ML}}$ of θ. Providing appropriate regularity conditions, show that $\hat{\theta}_{\texttt{1step}}$ and $\hat{\theta}_{\texttt{ML}}$ are asymptotically equivalent, that is, $\sqrt{n}(\hat{\theta}_{\texttt{1step}} - \hat{\theta}_{\texttt{ML}}) = o_{\mathbb{P}}(1)$.

- **Solution**: Notice that $0 = \ell_{\theta,n}(\hat{\theta}_{\texttt{ML}}) = \ell_{\theta,n}(\theta) + \ell_{\theta\theta,n}(\bar{\theta})(\hat{\theta}_{\texttt{ML}} - \theta)$ for some $\bar{\theta}$ between $\hat{\theta}_{\texttt{ML}}$ and θ. It immediately follows that

$$\begin{aligned}\sqrt{n}(\hat{\theta}_{\texttt{1step}} - \hat{\theta}_{\texttt{ML}}) &= \sqrt{n}(\hat{\theta}_{\texttt{1step}} - \theta) - \sqrt{n}(\hat{\theta}_{\texttt{ML}} - \theta) \\ &= -\frac{\sqrt{n}\ell_{\theta,n}(\theta)}{\ell_{\theta\theta,n}(\check{\theta})} + \frac{\sqrt{n}\ell_{\theta,n}(\theta)}{\ell_{\theta\theta,n}(\bar{\theta})} + o_{\mathbb{P}}(1).\end{aligned}$$

Since $\sqrt{n}\ell_{\theta,n}(\theta) = O_{\mathbb{P}}(1)$ and both $\check{\theta}$ and $\bar{\theta}$ are consistent, under regularity conditions, $\sqrt{n}(\hat{\theta}_{\texttt{1step}} - \hat{\theta}_{\texttt{ML}}) = o_{\mathbb{P}}(1)$.

5. Consider now the special case $X_i \sim \mathsf{Cauchy}(\theta, 1)$, which has Lebesgue density

$$f(x-\theta) = \frac{1}{\pi}\frac{1}{1+(x-\theta)^2}.$$

(a) Show that $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i \sim \mathsf{Cauchy}(\theta, 1)$.
Is $\bar{X}_n$ a consistent estimator of θ?

- **Solution**: By the Cauchy characteristic function, $\mathbb{E}\left[e^{itX_i}\right] = e^{i\theta t-|t|}$. Therefore by independence, $\mathbb{E}\left[e^{it\bar{X}_n}\right] = \left(e^{i\theta t/n-|t/n|}\right)^n$ and $\bar{X}_n \sim \mathsf{Cauchy}(\theta, 1)$ also. Then $\bar{X}_n$ does not converge in probability.

(b) Show that $\ell_{\theta,n}(\theta)$ has at least one root.
Show that if θ^* is a root of $\ell_{\theta,n}(\theta)$, then $\theta^* \in [\min_i X_i, \max_i X_i]$ almost surely.

- **Solution**: The score is

$$\ell_{\theta,n}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \frac{2(X_i - \theta)}{1 + (X_i - \theta)^2}.$$

Note that for $\theta < \min_i X_i$, the numerator of each summand is positive, while the denominator is always at least 1. Similarly for $\theta > \max_i X_i$ the numerators are negative, and appeal to the intermediate value theorem.

(c) Find the asymptotic efficiency of $\hat{\theta}_{\texttt{1step}}$ relative to $\hat{\theta}_{(1/2)}$.
Hint: for all $u > 0$,

$$J(u) := \int_{\mathbb{R}} \frac{1}{u + x^2}\,\mathrm{d}x = \frac{\pi}{\sqrt{u}}.$$

- **Solution**: We have by part 2 that

$$\sqrt{n}(\hat{\theta}_{\texttt{1step}} - \theta) \rightsquigarrow \mathsf{N}(0, 2) \qquad \text{and} \qquad \sqrt{n}(\hat{\theta}_{(1/2)} - \theta) \rightsquigarrow \mathsf{N}(0, \pi^2/4),$$

because

$$\begin{aligned}
\mathcal{I}(\theta) &= \int_{\mathbb{R}} \frac{f'(x)^2}{f(x)}\,\mathrm{d}x \\
&= \frac{1}{\pi}\int_{\mathbb{R}} \frac{4x^2}{(1+x^2)^3}\,\mathrm{d}x \\
&= \frac{1}{\pi}\int_{\mathbb{R}} \left(\frac{4}{(1+x^2)^2} - \frac{4}{(1+x^2)^3}\right)\mathrm{d}x \\
&= \frac{1}{\pi}\left(-4J'(1) - 2J''(1)\right) \\
&= 2 - 3/2 \\
&= 1/2
\end{aligned}$$

and $4f(0)^2 = 4/\pi^2$. Thus, the asymptotic efficiency of $\hat{\theta}_{\texttt{1step}}$ relative to $\hat{\theta}_{(1/2)}$ is $8/\pi^2 \approx 0.81$.

(d) Let $\theta_0 > 0$ and consider the one-sided testing problem:

$$\mathsf{H}_0 : \theta = \theta_0 \qquad \text{vs.} \qquad \mathsf{H}_1 : \theta > \theta_0.$$

Propose an asymptotic level $\alpha \in (0, 1)$ hypothesis test based on $\hat{\theta}_{\texttt{1step}}$.
Show that the power function of your test satisfies $\lim_{n\to\infty} \beta_n(\theta) = 1$ for all $\theta > \theta_0$.

- **Solution**: We consider a Wald test based on the asymptotically efficient estimator $\hat{\theta}_{\texttt{1step}}$. Recall we have

$$T_{\texttt{W}}(\theta_0) = \sqrt{\frac{n}{2}}(\hat{\theta}_{\texttt{1step}} - \theta_0) \rightsquigarrow \mathsf{N}(0,1),$$

and therefore an asymptotic level $\alpha \in (0,1)$ test rejects H_0 if and only if $T_{\texttt{W}}(\theta_0) > \Phi^{-1}(1-\alpha)$. The power function is

$$\begin{aligned}\beta_n(\theta) &= \mathbb{P}_\theta\left[T_{\texttt{W}}(\theta_0) > \Phi^{-1}(1-\alpha)\right] \\ &= \mathbb{P}_\theta\left[\sqrt{\frac{n}{2}}(\hat{\theta}_{\texttt{1step}} - \theta) > \Phi^{-1}(1-\alpha) - \sqrt{\frac{n}{2}}(\theta - \theta_0)\right] \\ &= 1 - \Phi\left(\Phi^{-1}(1-\alpha) - \sqrt{\frac{n}{2}}(\theta-\theta_0)\right) + o(1).\end{aligned}$$

Therefore, it follows that $\lim_{n\to\infty}\beta_n(\theta) = 1$ for all $\theta > \theta_0$ (under H_1).

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m$, $x = a_1, ..., a_m$
	m.g.f.	$\sum_{j=1}^{m} e^{a_j t}/m$, $t \in \mathcal{R}$
$DU(a_1, ..., a_m)$	Expectation	$\sum_{j=1}^{m} a_j/m$
	Variance	$\sum_{j=1}^{m}(a_j - \bar{a})^2/m$, $\bar{a} = \sum_{j=1}^{m} a_j/m$
	Parameter	$a_i \in \mathcal{R}$, $m = 1, 2, ...$
Binomial	p.d.f.	$\binom{n}{x} p^x(1-p)^{n-x}$, $x = 0, 1, ..., n$
	m.g.f.	$(pe^t + 1 - p)^n$, $t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1]$, $n = 1, 2, ...$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!$, $x = 0, 1, 2, ...$
	m.g.f.	$e^{\theta(e^t - 1)}$, $t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p$, $x = 1, 2, ...$
	m.g.f.	$pe^t/[1 - (1-p)e^t]$, $t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-	p.d.f.	$\binom{n}{x}\binom{m}{r-x} / \binom{N}{r}$
geometric		$x = 0, 1, ..., \min\{r, n\}$, $r - x \leq m$
	m.g.f.	No explicit form
$HG(r, n, m)$	Expectation	rn/N
	Variance	$rnm(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, ...$, $N = n + m$
Negative	p.d.f.	$\binom{x-1}{r-1} p^r(1-p)^{x-r}$, $x = r, r+1, ...$
binomial	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r$, $t < -\log(1-p)$
	Expectation	r/p
$NB(p, r)$	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1]$, $r = 1, 2, ...$
Log-	p.d.f.	$-(\log p)^{-1}x^{-1}(1-p)^x$, $x = 1, 2, ...$
distribution	m.g.f.	$\log[1 - (1-p)e^t]/\log p$, $t \in \mathcal{R}$
	Expectation	$-(1-p)/(p \log p)$
$L(p)$	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt}-e^{at})/[(b-a)t],\ \ t\in\mathcal{R}$
$U(a,b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a,\ b\in\mathcal{R},\ \ a<b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t+\sigma^2t^2/2},\ \ t\in\mathcal{R}$
$N(\mu,\sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1},\ \ t<\theta^{-1}$
$E(a,\theta)$	Expectation	$\theta+a$
	Variance	θ^2
	Parameter	$\theta>0,\ \ a\in\mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2},\ \ t<1/2$
χ^2_k	Expectation	k
	Variance	$2k$
	Parameter	$k=1,2,...$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha},\ \ t<\gamma^{-1}$
$\Gamma(\alpha,\gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma>0,\ \alpha>0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha,\beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha>0,\ \ \beta>0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t-\sigma\|t\|}$
$C(\mu,\sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)}\left(1+\frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, $(n>1)$
	Variance	$n/(n-2)$, $(n>2)$
	Parameter	$n=1,2,...$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, $(m>2)$
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, $(m>4)$
	Parameter	$n=1,2,...,\ \ m=1,2,...$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x-\mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu,\sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2}-1)$
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha,\theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1}+1)$
	Variance	$\theta^{2/\alpha}\left\{\Gamma(2\alpha^{-1}+1)-\left[\Gamma(\alpha^{-1}+1)\right]^2\right\}$
	Parameter	$\theta>0,\ \ \alpha>0$
Double	p.d.f.	$\frac{1}{2\theta}e^{-\lvert x-\mu\rvert/\theta}$
Exponential	m.g.f.	$e^{\mu t}/(1-\theta^2t^2)$, $\lvert t\rvert<\theta^{-1}$
	Expectation	μ
$DE(\mu,\theta)$	Variance	$2\theta^2$
	Parameter	$\mu\in\mathcal{R},\ \ \theta>0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a,\theta)$	Expectation	$\theta a/(\theta-1)$, $(\theta>1)$
	Variance	$\theta a^2/[(\theta-1)^2(\theta-2)]$, $(\theta>2)$
	Parameter	$\theta>0,\ \ a>0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1+e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1+\sigma t)\Gamma(1-\sigma t)$, $\lvert t\rvert<\sigma^{-1}$
$LG(\mu,\sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu\in\mathcal{R}$, $\sigma>0$